

Varun

• Data Science • Machine Learning • Computational Biology
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SUMMARY

BS-MS graduate (IISER Pune, May 2026) with hands-on experience building end-to-end data and ML pipelines for complex biological and image datasets. Proficient in Python, PyTorch, and scikit-learn, with practical exposure to regression, tree-based models (Random Forest, XGBoost), CNNs, and NLP. Experienced in the full data workflow, from preprocessing and feature engineering to model validation, benchmarking, and reporting. Thesis work at IIIT Delhi under Prof. G.P.S. Raghava (Director, IIIT Delhi) involved large-scale structural benchmarking across multiple datasets, precision-condition comparisons, and automated batch processing pipelines. Comfortable working with messy, real-world data and translating technical findings into clear, actionable insights.

SKILLS

Languages: Python, R (basic), SQL, SAS (basic), JavaScript, HTML, CSS

ML / DL Frameworks: PyTorch, TensorFlow, scikit-learn, CNN, NLP, Few-Shot Learning

Bioinformatics Tools: AlphaFold 2, BLAST, Biopython, MACE-OFF23, ET-Flow, OpenCV

Data & Visualization: Pandas, NumPy, Excel, Tableau, Power BI

Dev Tools: Git, Anaconda, Google Colab

Soft Skills: Analytical Thinking, Problem-Solving, Communication, Leadership

PUBLICATIONS / PREPRINTS

“PEPstrMOD2: Next-generation tertiary structure prediction of chemically modified and non-natural peptides.” bioRxiv (2026). DOI: [10.64898/2026.06.22.733733](https://doi.org/10.64898/2026.06.22.733733)

EXPERIENCE

Master's Thesis – IIIT Delhi, Computational Biology

Supervisor: Prof.G.P.S.Raghava(Director,IIITDelhi) | May 2025 – Apr 2026

Alpha-Mod: End-to-end pipeline for tertiary structure prediction of chemically modified peptides

- Built an end-to-end ML pipeline combining Alpha fold 2, a large language model (ESM Fold), a generative flow model, and a physics-based optimization stage to predict 3D molecular structures from sequence inputs
- Designed and ran a systematic benchmarking study across multiple datasets and model configurations (precision settings, convergence thresholds); evaluated outputs using domain-standard quality metrics and geometric similarity scores
- Automated batch processing, parallel execution, failure detection, and result aggregation in Python; built resume logic to handle large-scale interrupted runs reliably

GitHub Repository: [Alpha-Mod](#)

Image & Video Processing Project – IISER Pune

Jan 2025 – Apr 2025

Skin Disease Classification Using AI

- Designed a CNN-based deep learning pipeline for multi-class skin disease image classification, applying data augmentation and transfer learning techniques to improve model generalization.
- Built real-time object detection and video anomaly detection models using OpenCV and TensorFlow; implemented detection, segmentation, and tracking workflows.
- Automated the full preprocessing-to-evaluation pipeline for reproducibility and delivered structured technical reports documenting model performance and findings.

GitHub Repository: [project_Varun](#)

Semester Project – M.S. Madhusudhan Lab (COSPI), IISER Pune

Bioinformatics & Computational Biology

- Built a Python pipeline to parse, extract, and annotate sequence data from two large genomic datasets; validated outputs against reference databases using sequence alignment tools
- Automated the full data processing workflow end-to-end, improving reproducibility and generating structured summary reports from raw outputs

EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT Delhi)

Master's Thesis (Visiting) | May 2025 – Apr 2026

Thesis: Alpha-Mod - Predicting 3D structures of chemically modified peptides using attention-based and physics-based methods

Indian Institute of Science Education and Research (IISER), Pune

BS-MS Integrated Dual Degree (Biology) | 2021 – May 2026

Relevant Courses: Data Analysis, Statistical Learning & Data Science, Image & Video Processing with Deep Learning, Bioinformatics, Probability & Statistics

Modern School, Ghaziabad

10th (CBSE): 92.5% | 12th (CBSE): 93% | 2018–2020

AWARDS & RECOGNITION

- Ranked in Top 1 percentile in IISER Aptitude Test (IAT), 2021
- Recipient of School Merit Scholarships - Indian Post & Modern School (2018–2020)
- Silver Medalist, International Olympiad of Mathematics (IOM), Silver Zone, school level
- Gold Medalist, IISER Badminton League (IBL) - Men's Doubles (2023)

EXTRACURRICULARS

- Social Media Lead, PIXEL (Photography Club) - grew followers by 500% in 1 month
- Social Media Lead, EiC (Entrepreneurship & Innovation Club), IISER Pune
- Volunteer, Abhyasika - teaching underprivileged children